

KCET – Chemistry – 2018
VERSION CODE: A

Question 1: 1.0 g of Mg is burnt with 0.28 g of O₂ in a closed vessel. Which reactant is left in excess and how much?

- (A) Mg, 5.8 g
- (B) Mg, 0.58 g
- (C) O₂, 0.24 g
- (D) O₂, 2.4 g

Answer 1: (B)

Question 2: The orbital nearest to the nucleus is

- (A) 4f
- (B) 5d
- (C) 4s
- (D) 7p

Answer 2: (C)

Question 3: Which of the following is the correct order of radius?

- (A) $H > H > H^+$
- (B) $Na^+ > F^- > O^{2-}$
- (C) $F^- > O^{2-} > Na^+$
- (D) $Al^{3+} > Mg^{2+} > N^{3-}$

Answer 3: (A)

Question 4: The intramolecular hydrogen bond is present in

- (A) Phenol
- (B) o-Nitrophenol
- (C) p-Nitrophenol
- (D) p-Cresol

Answer 4: (B)

Question 5: The state of hybrid orbitals of carbon in CO₂, CH₄ and CO₃²⁻ respectively is

- (A) sp³, sp² and sp
- (B) sp³, sp and sp²
- (C) sp, sp³ and sp²
- (D) sp², sp³ and sp

Answer 5: (C)

Question 6: For an ideal gas, compressibility factor is

- (A) 0
- (B) 1
- (C) -1
- (D) +2

Answer 6: (B)

Question 7: The relationship between K_p and K_c is $K_p = K_c(RT)^{\Delta n}$. What would be the value of Δn for the reaction $\text{NH}_4\text{Cl(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{HCl (g)}$?

- (A) 1
- (B) 0.5
- (C) 1.5
- (D) 2

Answer 7: (D)

Question 8: Acidity of BF_3 can be explained on which of the following concepts?

- (A) Arrhenius concept
- (B) Bronsted Lowry concept
- (C) Lewis concept
- (D) Bronsted Lowry as well as Lewis concept

Answer 8: (C)

Question 9: For the redox reaction $x\text{MnO}_4^- + y\text{H}_2\text{C}_2\text{O}_4 + z\text{H}^+ \rightarrow m\text{Mn}^{2+} + n\text{CO}_2 + p\text{H}_2\text{O}$. The values of x, y, m and n are

- (A) 10, 2, 5, 2
- (B) 2, 5, 2, 10
- (C) 6, 4, 2, 4
- (D) 3, 5, 2, 10

Answer 9: (B)

Question 10: H_2O_2 is

- (A) An oxidising agent
- (B) A reducing agent
- (C) Both Oxidising and reducing agent
- (D) Neither oxidising nor reducing agent

Answer 10: (C)

Question 11: Dead burnt plaster is

- (A) CaSO_4
- (B) $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$
- (C) $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
- (D) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Answer 11: (A)

Question 12: Identify the following compound which exhibits geometrical isomerism:

- (A) But-2-ene
- (B) But-1-ene
- (C) Butane
- (D) Isobutane

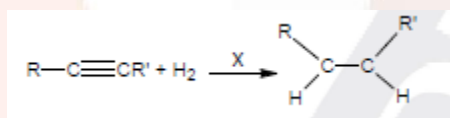
Answer 12: (A)

Question 13: During the fusion of organic compound with sodium metal, nitrogen present in the organic compound is converted into

- (A) NaNO_2
- (B) NaNH_2
- (C) NaCN
- (D) NaNC

Answer 13: (C)

Question 14: The reagent 'X' used for the following reaction is



- (A) Ni
- (B) Pd/C
- (C) LiAlH_4
- (D) Na/Liquid NH_3

Answer 14: (B)

Question 15: Which of the following ions will cause hardness in water?

- (A) Ca^{2+}
- (B) Na^+
- (C) Cl^-
- (D) K^+

Answer 15: (A)

Question 16: Which of the following oxides shows electrical properties like metals?

- (A) SiO_2
- (B) MgO
- (C) $\text{SO}_2(\text{s})$
- (D) CrO_2

Answer 16: (D)

Question 17: Which of the following aqueous solutions should have the highest boiling point?

- (A) 1.0 M NaOH
- (B) 1.0 M Na_2SO_4
- (C) 1.0 M NH_2NO_3
- (D) 1.0 M KNO_3

Answer 17: (B)

Question 18: The charge required for the reduction of 1 mole of MnO_4^- to MnO_2 is

- (A) 1F
- (B) 3F
- (C) 5F
- (D) 7F

Answer 18: (B)

Question 19: For the reaction, $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$, the rate of disappearance of O_2 is $2 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$.

The rate of appearance of SO_3 is

- (A) $2 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- (B) $4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- (C) $1 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- (D) $6 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

Answer 19: (B)

Question 20: Which of the following electrolytes will have maximum coagulating value for AgI/Ag^+ sol?

- (A) Na_2S
- (B) Na_3PO_4
- (C) Na_2SO_4
- (D) NaCl

Answer 20: (D)

Question 21: Electrolytic refining is used to purify which of the following metals?

- (A) Cu and Zn
- (B) Ge and Si
- (C) Zr and Ti
- (D) Zn and Hg

Answer 21: (A)

Question 22: Dry ice is

- (A) Solid CO
- (B) Solid SO₂
- (C) Solid CO₂
- (D) Solid O₂

Answer 22: (C)

Question 23: Which of the following is an amphoteric oxide?

- (A) V₂O₅, Cr₂O₃
- (B) Mn₂O₇, Cr₂O₃
- (C) CrO, V₂O₅
- (D) V₂O₅, V₂O₄

Answer 23: (A)

Question 24: The IUPAC name of [Co(NH₃)₄Cl(NO₂)]Cl is

- (A) tetraamminechloridonitrito-N-cobalt(III) chloride
- (B) tetraamminechloridonitrocobalt(II) chloride
- (C) tetraamminechloridonitrocobalt(I) chloride
- (D) tetraamminechloridodinitrocobalt(III) chloride

Answer 24: (A)

Question 25: Which of the following statements is true in case of alkyl halides?

- (A) They are polar in nature
- (B) They can form hydrogen bonds
- (C) They are highly soluble in water
- (D) They undergo addition reactions

Answer 25: (A)

Question 26: Phenol can be distinguished from ethanol by the reagent

- (A) Bromine water
- (B) Sodium metal
- (C) Iron metal
- (D) Chlorine water

Answer 26: (A)

Question 27: Which of the following compounds undergoes haloform reaction?

- (A) CH₃COCH₃
- (B) HCHO
- (C) CH₃CH₂Br
- (D) CH₃ – O – CH₃

Answer 27: (A)

Question 28: Which of the following will be the most stable diazonium salt (RN_2^+X^-)?

- (A) $\text{CH}_3\text{N}_2^+\text{X}^-$
- (B) $\text{C}_6\text{H}_5\text{N}_2^+\text{X}^-$
- (C) $\text{CH}_3\text{CH}_2\text{N}_2^+\text{X}^-$
- (D) $\text{C}_6\text{H}_5\text{CH}_2\text{N}_2^+\text{X}^-$

Answer 28: (B)

Question 29: Which of the following bases is not present in DNA?

- (A) Adenine
- (B) Guanine
- (C) Cytosine
- (D) Uracil

Answer 29: (D)

Question 30: Which one of the following is a polyamide polymer?

- (A) Terylene
- (B) Nylon-6,6
- (C) Buna-S
- (D) Bakelite

Answer 30: (B)

Question 31: In F.C.C. the unit cell is shared equally by how many unit cells?

- (A) 10
- (B) 8
- (C) 6
- (D) 2

Answer 31: (C)

Question 32: At a particular temperature, the ratio of molar conductance to specific conductance of 0.01M NaCl solution is

- (A) $10^5 \text{ cm}^3 \text{ mol}^{-1}$
- (B) $10^3 \text{ cm}^3 \text{ mol}^{-1}$
- (C) $10 \text{ cm}^3 \text{ mol}^{-1}$
- (D) $10^5 \text{ cm}^2 \text{ mol}^{-1}$

Answer 32: (A)

Question 33: Isotonic solutions are solutions having the same

- (A) Surface tension
- (B) Vapour pressure
- (C) Osmotic pressure
- (D) Viscosity

Answer 33: (C)

Question 34: The temperature coefficient of a reaction is 2. When the temperature is increased from 30°C to 90°C, the rate of reaction is increased by

- (A) 150 times
- (B) 410 times
- (C) 72 times
- (D) 64 times

Answer 34: (D)

Question 35: Gold sol is not a

- (A) Lyophobic sol
- (B) Negatively charged sol
- (C) Macromolecular sol
- (D) Multimolecular colloid

Answer 35: (C)

Question 36: The common impurity present in bauxite is

- (A) CuO
- (B) ZnO
- (C) Fe₂O₃
- (D) Cr₂O₃

Answer 36: (C)

Question 37: Very pure N₂ can be obtained by

- (A) Thermal decomposition of ammonium dichromate
- (B) Treating aqueous solution of NH₄Cl and NaNO₂
- (C) Liquifaction and fractional distillation of liquid air
- (D) Thermal decomposition of sodium azide

Answer 37: (D)

Question 38: Which of the following oxidation states is common for all lanthanides?

- (A) +2
- (B) +3
- (C) +4

(D) +5

Answer 38: (B)

Question 39: The electronic configuration of transition element “X”, is +3, oxidation state is $[\text{Ar}]3d^5$. What is its atomic number?

(A) 25

(B) 26

(C) 27

(D) 24

Answer 39: (B)

Question 40: n-Propyl chloride reacts with sodium metal in dry ether to give

(A) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

(B) $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

(C) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

(D) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

Answer 40: (A)

Question 41: When the vapours of tertiary butyl alcohol are passed through heated copper at 573 K, the product formed is

(A) But-2-ene

(B) 2-Butanone

(C) 2-Methyl propene

(D) Butanal

Answer 41: (C)

Question 42: What is the increasing order of acidic strength among the following?

(i) p-methoxy phenol

(ii) p-methyl phenol

(iii) p-nitro phenol

(A) ii < iii < I

(B) iii < ii < I

(C) I < ii < iii

(D) I < iii < ii

Answer 42: (C)

Question 43: Which of the following is more basic than aniline?

- (A) Diphenylamine
- (B) Triphenylamine
- (C) p-nitroaniline
- (D) Benzylamine

Answer 43: (D)

Question 44: The two forms of D-Glucopyranose are called

- (A) Diastereomers
- (B) Anomers
- (C) Epimers
- (D) Enantiomers

Answer 44 : (B)

Question 45: Among the following, the branched chain polymer is

- (A) Polyvinyl chloride
- (B) Bakelite
- (C) Low density polythene
- (D) High density polythene

Answer 45: (C)

Question 46: Edge length of a cube is 300 pm. Its body diagonal would be

- (A) 600 pm
- (B) 423 pm
- (C) 519.6 pm
- (D) 450.5 pm

Answer 46: (C)

Question 47: Which of the following is not a conductor of electricity?

- (A) Solid NaCl
- (B) Cu
- (C) Fused NaCl
- (D) Brine solution

Answer 47: (A)

Question 48: For a cell involving two electron changes, $E_{cell}^0 = 0.3V$ at $25^\circ C$. The equilibrium constant of the reaction is

- (A) 10^{-10}
- (B) 3×10^{-2}
- (C) 10

(D) 10^{10}

Answer 48: (D)

Question 49: The value of rate constant of a pseudo first order reaction

- (A) Depends only on temperature
- (B) Depends on the concentration of reactants present in small amounts
- (C) Depends on the concentration of reactants present in excess
- (D) Is independent of the concentration of reactants

Answer 49: (A)

Question 50: $(\text{CH}_3)_3\text{SiCl}$ is used during polymerization of organosilicons because

- (A) The chain length of organosilicon polymers can be controlled by adding $(\text{CH}_3)_3\text{SiCl}$
- (B) $(\text{CH}_3)_3\text{SiCl}$ improves the quality and yield of the polymer
- (C) $(\text{CH}_3)_3\text{SiCl}$ does not block the end terminal of silicone polymer
- (D) $(\text{CH}_3)_3\text{SiCl}$ acts as a catalyst during polymerization.

Answer 50: (A)

Question 51: When PbO_2 reacts with concentrated HNO_3 , the gas evolved is

- (A) NO_2
- (B) O_2
- (C) N_2
- (D) N_2O

Answer 51: (B)

Question 52: KMnO_4 acts as an oxidising agent in alkaline medium. When alkaline KMnO_4 is treated with KI , iodide ion is oxidised to

- (A) I_2
- (B) IO^-
- (C) IO_3^-
- (D) IO_4^-

Answer 52: (C)

Question 53: $[\text{Fe}(\text{NO}_2)_3\text{Cl}_3]$ and $[\text{Fe}(\text{O} - \text{NO})_3\text{Cl}_3]$ shows

- (A) Linkage isomerism
- (B) Geometrical isomerism
- (C) Optical isomerism
- (D) Hydrate isomerism

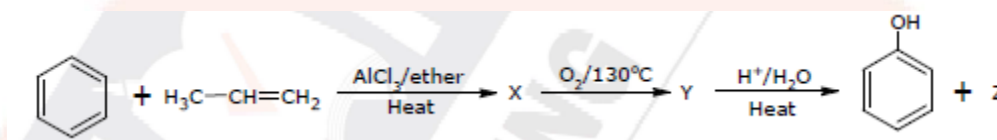
Answer 53: (A)

Question 54: Tertiary alkyl halide is practically inert to substitution by S_N2 mechanism because of

- (A) Insolubility
- (B) Instability
- (C) Inductive effect
- (D) Steric hindrance

Answer 54: (D)

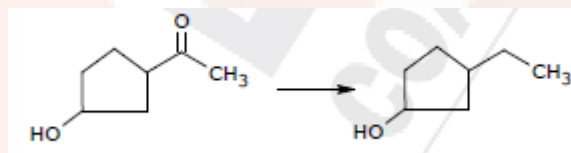
Question 55: The products X and Z in the following reaction sequence are



- (A) Isopropylbenzene and acetone
- (B) Cumene peroxide and acetone
- (C) Isopropylbenzene and isopropyl alcohol
- (D) Phenol and acetone

Answer 55: (A)

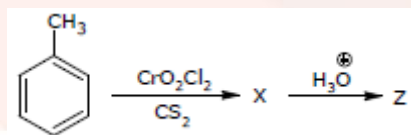
Question 56: The appropriate reagent for the following transformation is



- (A) $\text{Zn} - \text{Hg}/\text{HCl}$
- (B) $\text{H}_2\text{N} - \text{NH}_2$, KOH /ethylene glycol
- (C) Ni/H_2
- (D) NaBH_4

Answer 56: (B)

Question 57: In the following reaction,



The compound Z is

- (A) Benzoic acid
- (B) Benzaldehyde
- (C) Acetophenone
- (D) Benzene

Answer 57: (B)

Question 58: The reaction of Benzenediazonium chloride with aniline yields yellow dye. The name of the yellow dye is

- (A) p-Hydroxyazobenzene
- (B) p-Aminoazobenzene
- (C) p-Nitroazobenzene
- (D) o-Nitroazobenzene

Answer 58: (B)

Question 59: The glycosidic linkage involved in linking the glucose units in amylase part of starch is

- (A) $C_1 - C_4$ β -linkage
- (B) $C_1 - C_6$ α -linkage
- (C) $C_1 - C_6$ β -linkage
- (D) $C_1 - C_4$ α -linkage

Answer 59: (D)

Question 60: Ziegler-Natta catalyst is used to prepare

- (A) Low-density polythene
- (B) Teflon
- (C) High density polythene
- (D) Nylon-6

Answer 60: (C)